



Fig. 3: The IoT and AR in Volvo assembly line



Fig. 5: Telepresence in Industry 4.0

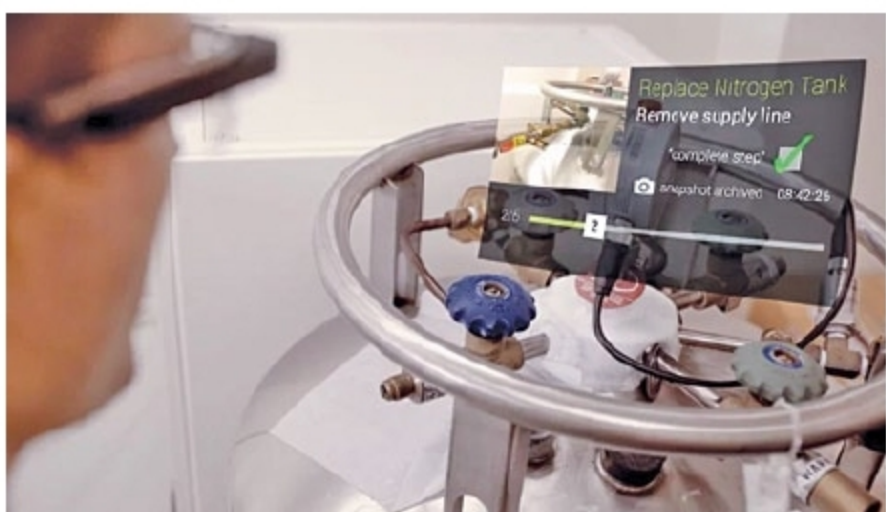


Fig. 4: AR display for maintenance in Industry 4.0



Fig. 6: AR and the IoT in Porsche factory

need to secure and protect industrial data processing systems from hackers and other cyber security threats. Reliable and sophisticated communication protocols help maintain secure communication between electronic devices.

Augmented reality. Playing a vital role in Industry 4.0, AR helps select manufacturing parts and send industrial fault status information to applications installed on users' phones. Many companies are developing applications to provide real-time information about product processing to workers. They are also making smart applications that help with decision making and improved work procedures.

Big data. In Industry 4.0, collection and analysis of data from different sources is a major problem. Big data can help optimise data, production quality, save energy and improve services. The main goal is to allow real-time decision making.

Autonomous robots. These robots handle complex tasks that are difficult

for human beings, by interacting with one another and with humans. These also reduce setup times and quality.

Additive manufacturing. In Industry 4.0, companies are adopting 3D printing technology for manufacturing and prototyping of products. 3D printing helps produce small batches of customised, high-performance products. This reduces transportation and management costs.

Simulation. 3D simulation is used in plant operations to monitor and analyse real-time data and create a mirror image of the physical world in a virtual model. This includes machines, products, sensors and humans. It helps drive down machine setup time and improve quality.

System integration. These days, information systems are not fully connected and integrated. Suppliers and customers are not synchronised with an organisation. In Industry 4.0, the whole organisation is inter- and intra-connected with each other.

Cloud computing. Industry 4.0 requires more data, storage and data

sharing across machines and data-limited boundaries. With advanced cloud computing, more machines and servers are deployed on the cloud. This helps attain better results and provide visibility to the entire supply chain.

The IoT. The IoT is defined as a combination of networks that enables dual communication between machinery, facilities, finished and unfinished products via local intelligence. It incorporates deep learning and machine learning, big data, industrial data sensors, machine-to-machine communication and more. Results, when compared to humans, are more accurate and reliable.

Applications of IoT and AR in Industry 4.0

Complex assembly. There are thousands of components that require expert skill-sets and instructions. A combination of the IoT and AR provides a solution for Industry 4.0. Recently, Volvo experimented with AR on its assembly line using Microsoft HoloLens (Fig. 3).